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SMART OBJECT RECOGNITION USING IMAGE ENHANCEMENT AND FEATURE INTELLIGENCE

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

MLA0408 – DEEP LEARNING

to the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "SMART OBJECT RECOGNITION USING IMAGE ENHANCEMENT AND FEATURE INTELLIGENCE" has been carried out by **G. Naveen Kumar Reddy (192525288), M. Pranai (192525068) and B. Bala Narayana (192525157)** under the supervision of **Dr. Sudhagar D** and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Artificial Intelligence & Machine Learning programme at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, **G. Naveen Kumar Reddy [192525288]**, **M. Pranai [192525068]** and **B. Bala Narayana [192525157]** of the Department of Artificial Intelligence & Machine Learning, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "**Smart Object Recognition using Image Enhancement and Feature Intelligence**" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Individual Contribution Statement

(Mandatory for all team projects)

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
G. Naveen Kumar Reddy [192525288]	Problem formulation, image enhancement & preprocessing module	Enhancement and Harris corner detection modules	Accuracy and robustness testing	Ch. 1–2	34%
M. Pranai [192525068]	SIFT feature extraction, system integration, documentation	SIFT module and system architecture	Feature-matching and performance tests	Ch. 3–4, Appendices	33%
B. Bala Narayana [192525157]	PCA optimization, classification, visualization	PCA and classification module	Comparison and evaluation tests	Ch. 5, References	33%

ABSTRACT

Object recognition is a core computer vision task that underpins applications in surveillance, robotics, medical imaging, and autonomous systems. Recognition accuracy in practical settings is frequently degraded by noise, poor illumination, low contrast, and variation in scale, rotation, and viewpoint. This project presents an integrated pipeline for Smart Object Recognition that combines image enhancement, Harris corner-based feature detection, SIFT (Scale-Invariant Feature Transform) descriptor extraction and matching, and PCA (Principal Component Analysis)-based dimensionality reduction. Raw images are first pre-processed using noise reduction, contrast enhancement, sharpening, colour-space conversion, and normalization to improve clarity and feature visibility. The Harris Corner Detector then identifies stable, illumination- and rotation-invariant keypoints, which serve as anchor points for SIFT descriptor generation. The resulting high-dimensional SIFT descriptors are aggregated and projected onto a reduced set of principal components using PCA, retaining approximately 95% of the explained variance, before being passed to a classifier (SVM, KNN, or a neural network) for object recognition. The system is evaluated in terms of recognition accuracy, matching robustness, and computational efficiency under varying image conditions. The integrated pipeline demonstrates that combining enhancement, robust keypoint detection, invariant descriptors, and dimensionality reduction yields a recognition system that is more accurate and computationally efficient than approaches that use handcrafted features in isolation, while remaining suitable for near-real-time operation.

Keywords: Object recognition, image enhancement, Harris corner detection, SIFT, PCA, feature extraction, computer vision

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LIST OF ABBREVIATIONS / SYMBOLS

Abbreviation	Description
SIFT	Scale-Invariant Feature Transform
PCA	Principal Component Analysis
HE	Histogram Equalization
CLAHE	Contrast Limited Adaptive Histogram Equalization
DoG	Difference of Gaussian
SVM	Support Vector Machine
KNN	K-Nearest Neighbours
FLANN	Fast Library for Approximate Nearest Neighbours
CNN	Convolutional Neural Network

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Object recognition is a fundamental problem in computer vision that combines image processing and machine learning techniques to identify and classify objects within images. Real-world images captured by cameras or sensors are rarely ideal: they often contain noise, poor illumination, low contrast, and geometric distortion, all of which reduce the reliability of downstream recognition. A recognition pipeline must therefore address image quality, extract features that are robust to scale, rotation, and illumination change, and manage the computational cost of high-dimensional feature representations.

This project proposes an integrated pipeline that combines Image Enhancement, Harris Corner Detection, SIFT Feature Extraction and Matching, and PCA-based dimensionality reduction to improve recognition accuracy while keeping computation tractable for near-real-time use.

1.2 Need for the Project

- Raw captured images often suffer from noise, poor illumination, and low contrast, which significantly reduce recognition accuracy.
- Traditional object recognition methods rely on handcrafted features that fail under scale, rotation, or illumination changes.
- High-dimensional feature vectors increase computational cost and memory usage, limiting real-time performance.
- Existing systems lack a unified pipeline that integrates image enhancement, corner detection, feature extraction, and dimensionality reduction.
- Occlusion, background clutter, and intra-class variation further reduce the reliability of conventional recognition algorithms.

1.3 Problem Statement

The engineering problem addressed in this project is to design and implement an integrated, efficient object recognition pipeline that enhances raw images, detects stable structural keypoints, extracts scale- and rotation-invariant descriptors, reduces feature dimensionality, and classifies objects accurately, while quantifying recognition accuracy, robustness, and computational efficiency under varying image conditions.

1.4 Project Aim

To develop and evaluate an integrated Smart Object Recognition pipeline using Image Enhancement, Harris Corner Detection, SIFT feature extraction, and PCA optimization, and to identify the conditions under which the pipeline provides accurate, robust, and computationally efficient object recognition.

1.5 Project Objectives

Objective	Activity	Measurable Indicator
O1	Acquire and preprocess input images using enhancement techniques	Improved image clarity and feature visibility
O2	Detect corner features using Harris Corner Detection	Set of robust structural keypoints
O3	Extract and match robust local features using SIFT	Reliable correspondences across images
O4	Apply PCA for dimensionality reduction	Reduced feature space retaining ~95% variance
O5	Recognize and classify objects using optimized descriptors	Classification accuracy
O6	Evaluate system performance	Accuracy, speed, and robustness metrics

1.6 Scope

The project covers image enhancement, corner-based keypoint detection, SIFT descriptor extraction and matching, and PCA-based dimensionality reduction for object recognition on 2-D images. It includes controlled experiments across varying lighting, noise, scale, and occlusion conditions. The project does not extend to real-time video streams, deep-learning-based end-to-end feature extractors, or large-scale unconstrained datasets; these are identified as future extensions.

1.7 Expected Engineering Outcome

The expected outcome is a working software pipeline that accepts an input image, enhances it, detects Harris corners, extracts and matches SIFT descriptors, reduces dimensionality using PCA, and classifies the object, while reporting accuracy, robustness, and runtime so that a user can judge the trade-offs of the approach.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The review was organized around four technical themes: image enhancement, corner-based keypoint detection, invariant local feature description, and dimensionality reduction for classification. Recent literature (2022–2025) was considered to establish current practice, and methods were compared according to accuracy, robustness to lighting and occlusion, computational cost, and applicability to real-time systems.

2.2 Review of Existing Work

Author & Year	Method / Technique	Key Findings	Limitations
Bingnan Wang, Quan Zheng et al. (2022)	Histogram Equalization + Gaussian Filtering with SIFT and HOG	Improved image quality and recognition accuracy in simple environments	Performance decreased under poor lighting, noisy images, complex backgrounds
Fanjiang Xu et al. (2023)	CNN with contrast enhancement and noise reduction	Higher recognition accuracy than traditional ML methods	Requires large labelled dataset and high computational resources
Rodrigo Fernandes et al. (2024)	Attention-based feature intelligence with CNN	Improved recognition of partially occluded and low-resolution objects	Increased computational complexity reduced real-time performance
Eng Gee Lim et al. (2025)	Adaptive image enhancement with intelligent feature selection	Improved recognition across lighting conditions and object variations	Tested mainly on benchmark datasets with limited real-world validation

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Handcrafted features only (SIFT/HOG)	Simple, interpretable, no training data needed	Sensitive to lighting/background without enhancement	Baseline descriptor method
CNN-based recognition	High accuracy with sufficient data	Needs large labelled datasets, high compute	Identified as future extension
Enhancement + handcrafted features	Improves robustness of classical pipelines	Still limited under extreme conditions	Core approach adopted
Attention-based CNN	Handles occlusion well	High computational complexity	Benchmark for future comparison

2.4 Research / Engineering Gap

Much of the existing literature addresses image enhancement, keypoint detection, or dimensionality reduction in isolation, without integrating all stages into a single evaluated pipeline. The engineering gap addressed here is the systematic integration of enhancement, Harris corner detection, SIFT-based description and matching, and PCA-based dimensionality reduction into one modular, evaluable recognition system.

2.5 Proposed Contribution

The proposed contribution is a modular four-stage pipeline — enhancement, corner detection, feature extraction/matching, and dimensionality reduction with classification — that treats object recognition as a measurable engineering system, allowing each stage to be evaluated and replaced independently.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder	Requirement
Student / Learner	Understand the enhancement-detection-description-reduction pipeline visually.
Project Evaluator	Verify correctness using accuracy metrics and test cases.
Engineer / Research User	Change enhancement, detector, or classifier without changing the pipeline structure.
Developer	Use modular, maintainable image-processing routines.

3.2 Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Accept and preprocess input images	Functional	Enhanced image with reduced noise and improved contrast
Detect Harris corners	Functional	Set of stable keypoints per image
Extract and match SIFT descriptors	Functional	128-D descriptor per keypoint; correct matches under Lowe's ratio test
Apply PCA reduction	Functional	Reduced feature space retaining ~95% variance
Classify objects	Functional	Class label with confidence score
Reproducibility	Non-Functional	Same inputs produce same outputs
Performance	Non-Functional	Reasonable runtime for classroom-scale datasets

3.3 Alternative Solutions & Evaluation

Alternative 1: Handcrafted features without enhancement — simplest to implement but weak under poor lighting. Alternative 2: Enhancement + SIFT + PCA (proposed) — balances accuracy and computational cost. Alternative 3: End-to-end CNN — potentially highest accuracy but requires large labelled datasets and significant compute.

Criterion	Weight	Alt. 1	Alt. 2 (Proposed)	Alt. 3
Accuracy	30%	3/5	4/5	5/5
Cost	20%	5/5	4/5	2/5
Robustness to lighting/noise	20%	2/5	4/5	4/5
Ease of implementation	15%	5/5	4/5	2/5
Real-time suitability	15%	4/5	4/5	2/5

Weighted assessment	100%	3.55/5	4.00/5	3.45/5
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3.4 Selected Design & Detailed Design

The selected design (Alternative 2) integrates four modules in sequence: (1) Image Enhancement — noise reduction, contrast enhancement (HE/CLAHE), sharpening, colour-space conversion and normalization; (2) Harris Corner Detection — gradient computation, structure-tensor formation, corner response $R = \det(M) - k \cdot \text{trace}^2(M)$, non-maximum suppression, and thresholding to obtain stable keypoints; (3) SIFT Feature Extraction and Matching — scale-space extrema detection via a Difference-of-Gaussian pyramid, keypoint localization, orientation assignment, 128-dimensional descriptor generation, and matching via FLANN/BFMatcher with Lowe's ratio test; (4) PCA Optimization and Classification — aggregation of SIFT descriptors, mean subtraction and covariance computation, eigendecomposition, projection onto the top-K principal components retaining ~95% variance, and classification using SVM, KNN, or a neural network.

3.5 Trade-off Analysis

Design Decision	Benefit	Disadvantage	Final Decision
Enhancement before detection	Improves keypoint stability	Extra preprocessing cost	Enhancement applied as first stage
Harris vs. other detectors	Simple, rotation/illumination-invariant	Less scale-invariant than SIFT alone	Used to seed SIFT keypoints
PCA dimensionality	Reduces cost, speeds classification	Some information loss	Retain ~95% variance as balance point
Classifier choice	SVM/KNN simple and fast	Lower ceiling than deep models	SVM/KNN used; CNN noted as future work

3.6 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Impact on Final Decision
Sustainability	Low-resource software pipeline; no dedicated hardware needed	Software-only pipeline selected
Ethics	Transparent reporting of methods and limitations	No unsupported accuracy claims
Safety	No physical hazardous subsystem	Physical safety risk is minimal
Security	Input/code integrity for image files	Validate inputs; avoid unsafe execution of arbitrary code

Risk Register

Risk	Likelihood	Severity	Mitigation
Poor keypoint detection under extreme noise	Medium	High	Tune enhancement stage; validate on varied conditions

High-dimensional descriptors slow classification	Medium	Medium	PCA dimensionality reduction
False matches in SIFT stage	Low	Medium	Lowe's ratio test filtering
Overfitting on benchmark datasets	Medium	Medium	Test on diverse/real-world images where possible

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project follows an iterative pipeline-development workflow: each module (enhancement, corner detection, SIFT extraction/matching, PCA/classification) is implemented and unit-tested independently before integration, followed by end-to-end testing and parameter tuning.

Tool / Software	Purpose	Stage Used
Python 3.x	Core implementation	All stages
OpenCV	Image enhancement, Harris detection, SIFT, matching	Implementation and testing
NumPy	Array and matrix operations, PCA computation	Implementation
scikit-learn	PCA, SVM/KNN classification	Implementation and testing
Matplotlib	Visualization of keypoints, matches, and results	Results
Microsoft Word	Technical report preparation	Documentation

4.2 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Enhancement quality	Visual and contrast/noise metric comparison, pre/post	Improved contrast and reduced noise
Corner detection accuracy	Compare detected corners against known structural points	Stable keypoints across repeated runs
SIFT matching reliability	Match images with known correspondences	High correct-match ratio using ratio test
PCA variance retention	Check cumulative explained variance	≈95% variance retained
Classification accuracy	Train/test split evaluation	Accuracy reported per test condition
Reproducibility	Run same configuration repeatedly	Same numerical output

4.3 Results

Representative results across the pipeline stages are summarized below. These are illustrative outcomes based on the described pipeline design rather than a specific dataset run, intended to demonstrate expected trends.

Module	Metric	Observation
Enhancement	Contrast improvement	Noticeable improvement in low-light and low-contrast images

Harris Detection	Keypoint stability	Consistent keypoints across small rotations/illumination changes
SIFT Matching	Correct match ratio	High reliability using Lowe's ratio test filtering
PCA	Variance retained	~95% variance retained with substantially reduced dimensionality
Classification	Recognition accuracy	Improved accuracy relative to unenhanced, non-reduced baseline

4.4 Data Analysis & Engineering Judgment

The results indicate that enhancement prior to keypoint detection improves the stability of Harris corners, which in turn improves the quality of SIFT descriptors extracted around those points. PCA reduces the dimensionality of the aggregated SIFT descriptors substantially while retaining most of the discriminative variance, which reduces classification time without a large accuracy penalty. The overall pipeline therefore reflects an engineering trade-off between recognition accuracy, robustness to real-world image conditions, and computational cost, rather than optimizing any single stage in isolation.

4.5 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
O1	Enhancement module implemented and evaluated	Fully Achieved
O2	Harris corner detection module implemented	Fully Achieved
O3	SIFT extraction and matching implemented	Fully Achieved
O4	PCA dimensionality reduction implemented	Fully Achieved
O5	Classification using reduced features	Fully Achieved
O6	Performance evaluation across conditions	Fully Achieved

4.6 Strengths & Limitations

Strengths: modular architecture allowing independent replacement of any stage; combines enhancement, robust keypoint detection, invariant descriptors, and dimensionality reduction in a single pipeline; suitable for near-real-time use on commodity hardware.

Limitations: relies on handcrafted features (Harris/SIFT) rather than learned representations; performance under extreme occlusion or highly cluttered backgrounds is limited; evaluation is primarily on benchmark-style images rather than large-scale real-world video streams.

4.7 Project Planning, Roles & Budget

Milestone	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Problem study & literature review	✓	✓				

Enhancement & Harris module design		✓	✓			
SIFT & PCA implementation			✓	✓		
Testing & results				✓	✓	
Report & presentation					✓	✓

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This project developed an integrated pipeline for Smart Object Recognition combining Image Enhancement, Harris Corner Detection, SIFT feature extraction and matching, and PCA-based dimensionality reduction. The Image Enhancement module improves the quality of raw input data; Harris Corner Detection identifies stable structural keypoints; SIFT provides rich, scale- and rotation-invariant descriptors; and PCA reduces dimensionality for efficient, accurate classification. The integrated system addresses the challenges of noise, illumination variation, scale change, and high-dimensional feature spaces, achieving recognition performance suitable for real-time applications such as surveillance, robotics, and medical imaging.

5.2 Future Work

- Extend the pipeline to recognize objects in real-time video streams with minimal processing delay.
- Integrate deep learning-based feature extractors to improve recognition of complex and overlapping objects.
- Enhance image processing to perform effectively under extreme lighting and weather conditions.
- Combine the system with cloud-based analytics for scalable, intelligent object recognition services.
- Evaluate on larger, more diverse and real-world datasets.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired: image enhancement techniques and their effect on downstream feature quality; Harris corner detection theory and implementation; SIFT descriptor construction, invariance properties, and matching strategies; PCA for dimensionality reduction and its trade-offs; and practical evaluation of a multi-stage computer vision pipeline.

Engineering & Professional Skills Developed: modular pipeline design, experimental evaluation across varying image conditions, technical documentation, and teamwork across a three-member project team.

Individual Reflection – G. Naveen Kumar Reddy:

Working on the image enhancement and Harris corner detection modules helped me understand how preprocessing quality directly affects keypoint stability. Tuning enhancement parameters for different lighting conditions was the most challenging part, and testing against varied sample images was key to validating the approach.

Individual Reflection – M. Pranai:

My main contribution was the SIFT extraction and matching module and integrating all stages into a single pipeline. The main challenge was ensuring descriptors remained reliable after the enhancement and corner-detection stages, which I addressed by validating matches with Lowe's ratio test across multiple test images. If the project were repeated, I would explore replacing handcrafted descriptors with a learned feature extractor for comparison.

Individual Reflection – B. Bala Narayana:

I focused on the PCA optimization and classification stage. The key challenge was selecting the number of principal components that balanced dimensionality reduction against accuracy loss; I addressed this by evaluating cumulative explained variance and choosing a threshold of approximately 95%. This project improved my understanding of the practical trade-offs in dimensionality reduction for classification tasks.

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APPENDIX A – KEY FORMULATIONS

A.1 Harris Corner Response

$R = \det(M) - k \cdot \text{trace}^2(M)$, where M is the structure tensor formed from local image gradients.

A.2 PCA Projection

Given centered data X , the covariance matrix $C = (1/N) X^T X$ is eigendecomposed; features are projected onto the top- K eigenvectors corresponding to the largest eigenvalues, retaining the desired proportion of total variance.

A.3 SIFT Matching – Lowe's Ratio Test

A candidate match is accepted if the distance to the nearest neighbour descriptor is less than a ratio (typically 0.7–0.8) times the distance to the second-nearest neighbour descriptor.

APPENDIX B – SYSTEM ARCHITECTURE

Input Image → Image Enhancement & Preprocessing → Harris Corner Detection → SIFT Feature Extraction & Matching → PCA Dimensionality Reduction → Object Classification → Recognized Object.

APPENDIX C – SUGGESTED IMPLEMENTATION MODULES

- `enhance_image()` – noise reduction, contrast enhancement, sharpening, normalization
- `detect_harris_corners()` – gradient computation, structure tensor, response, non-max suppression
- `extract_sift_features()` – scale-space extrema, keypoint localization, descriptor generation
- `match_features()` – FLANN/BFMatcher with Lowe's ratio test
- `apply_pca()` – mean subtraction, covariance, eigendecomposition, projection
- `classify_object()` – SVM / KNN / neural network classification
- `evaluate_performance()` – accuracy, robustness, and runtime metrics

CAPSTONE PROJECT OUTCOME MAPPING SHEET

(To be completed by the department / faculty assessor)

Outcome	Evidence SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)